# Network Threats and Mitigation & Physical Security and Risk

**Q1. Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).**

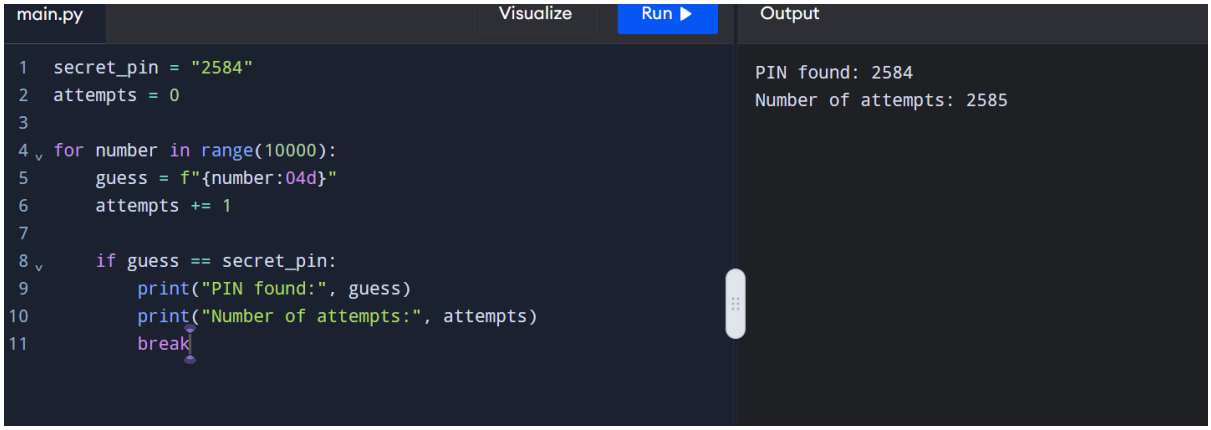
**Solution : Five common network threats with real-world examples**

| No. | Network Threat | Real-World Example | Mitigation |
|---|---|---|---|
| 1 | Phishing | A fake Instagram message asks a user to enter their password on a fraudulent login page. | Verify links and enable MFA. |
| 2 | DDoS Attack | Attackers flood a ticket-booking website with traffic, making it unavailable to genuine users. | Use DDoS protection and traffic filtering. |
| 3 | Man-in-the-Middle (MITM) | An attacker on an insecure public Wi-Fi network attempts to intercept a user's communication. | Use HTTPS and a trusted VPN. |
| 4 | Malware | A malicious attachment infects a computer and attempts to steal saved passwords. | Use updated antivirus software and avoid suspicious attachments. |
| 5 | Password Attack | An attacker tries commonly used passwords to gain access to a Paytm or email account. | Use strong, unique passwords, MFA, and login rate limits. |

**Q2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value. Constraint : The script should print the number of attempts it took to guess the correct PIN.**

**Solution : Python script to simulate a brute-force attack**

This program checks every possible four-digit PIN, from 0000 to 9999, against a preset PIN and prints the number of attempts needed.



**Q3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.**

**Solution : Four physical security risks in a server room**

| No. | Physical Risk | Mitigation |
|---|---|---|
| 1 | Unauthorized entry | Use biometric access or access cards. |
| 2 | Fire | Install smoke detectors and appropriate fire-suppression systems. |
| 3 | Power failure | Use UPS systems and backup power. |
| 4 | Overheating | Install air conditioning and temperature monitoring. |

**Q4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).**

**Solution : Small office floor plan with physical security measures**



**Q5. Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.**

**Solution : Seven-point security checklist for a college computer lab**

| No. | Action | Purpose |
|---|---|---|
| 1 | Install and regularly update antivirus software. | Protect computers from malware. |
| 2 | Configure firewalls on lab computers and network devices. | Block unauthorized network connections. |
| 3 | Use strong passwords and individual student accounts. | Prevent unauthorized account access. |
| 4 | Enable MFA for administrator accounts where supported. | Add another layer of authentication. |
| 5 | Install CCTV and restrict access to the network/server room. | Protect physical equipment. |
| 6 | Secure routers, switches, and network cables. | Prevent tampering and theft. |
| 7 | Back up important lab data and regularly install security updates. | Recover from failures and fix known vulnerabilities. |